

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2018

Saturday 26th May 2018, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;

in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy in which each firm can produce output X using labour input L according to a production function $X = AL^\alpha/\alpha$, where A is a technology parameter and $0 < \alpha < 1$. Suppose that the price of X is p , the price of L is w , and production is chosen by each firm to maximise profits.

- (a) Show that a firm's labour demand function is $L = (pA/w)^{1/(1-\alpha)}$.

Derive its output supply function, and its level of profits as functions of p and w .

- (b) What proportion of the value of output goes as profits, and what proportion as wages?

There are M capitalists, each of whom owns a single firm, and N workers, each endowed with a single unit of labour. All the capitalists and workers spend their entire income (from profits and wages respectively) on output, and their utility is equal to units of X consumed. Henceforward, take one unit of labour as the numeraire.

- (c) Show that the equilibrium value of p is given by $(pA)^{1/(1-\alpha)} = N/M$.

- (d) Technical progress improves the efficiency of the technology.

What are the equilibrium effects of such an increase in the parameter A ?

How are the benefits of the increase distributed between capitalists and workers?

[30%]

- (a) $\pi = pX - wL$, with $X = AL^\alpha/\alpha$, so $\pi = pAL^\alpha/\alpha - wL$

$$\pi' = pAL^{\alpha-1} - w, \text{ which } = 0 \text{ when } L^{\alpha-1} = w/pA,$$

$$\text{so } \boxed{L = (pA/w)^{1/(1-\alpha)}}, \quad \boxed{X = (A/\alpha)(pA/w)^{\alpha/(1-\alpha)}},$$

$$\pi = pX - wL = \frac{1}{\alpha} w(pA/w)^{1/(1-\alpha)} - w(pA/w)^{1/(1-\alpha)} = \boxed{\frac{1-\alpha}{\alpha} w(pA/w)^{1/(1-\alpha)}}$$

- (b) revenue = profits + wages

$$= \frac{1-\alpha}{\alpha} [w(pA/w)^{1/(1-\alpha)}] + [w(pA/w)^{1/(1-\alpha)}], \text{ so ratio is } \boxed{(1-\alpha) : \alpha}$$

- (c) $L_{\text{demand}} = M(pA/w)^{1/(1-\alpha)}$, $L_{\text{supply}} = N$, so in equilibrium $\boxed{(pA/w)^{1/(1-\alpha)} = N/M}$

- (d) If $A \uparrow$, then $p/w \downarrow$ so that pA/w is unchanged.

With $w = 1$ as the price of the numeraire, nominal wages and profits are unchanged; but $p \downarrow$, so real wages and profits increase (in same proportion).

Revenue shares of profits & wages unchanged at $(1-\alpha) : \alpha$.

2. What is the definition of a Nash equilibrium of a game?

Does the existence of a Nash equilibrium lead to a good prediction of how the players will play a game?

Consider the two-player, one-shot, simultaneous-move game (below), for $t = 2, 4, 3$.

[20%]

Nash equilibrium: a strategy profile $(s_1^*, s_2^*, \dots, s_K^*)$ such that s_k^* is a best response to s_{-k}^* for each player $k = 1, 2, \dots, K$.

If the game has a unique pure-strategy Nash equilibrium, then that could provide a good prediction of how the players might play.

However, with multiple equilibria and/or mixed-strategy equilibria, then that may well not lead to good prediction.

(a) When $t = 2$, column player has a strictly dominant strategy of *Right*:

	<i>Left</i>	<i>Right</i>
<i>Up</i>	2 0	3 3
<i>Down</i>	0 5	5 0

Row's best response to *Right* is *Up*, so NE is $(Up, Right)$.

(b) When $t = 4$, there is no NE in pure strategies:

	<i>Left</i>	<i>Right</i>
<i>Up</i>	4 0	3 3
<i>Down</i>	0 5	5 0

To make Column indifferent, Row plays *Up* w.p. q s.t. $4q = 3q + 5(1 - q)$, $q = 5/6$.

To make Row indifferent, Column plays *Right* w.p. p s.t. $3p = 5(1 - p)$, $p = 5/8$.

NE is $(Up \text{ w.p. } 5/6, Right \text{ w.p. } 5/8)$.

(c) When $t = 3$, if Row plays *Up* then Column is indifferent:

	<i>Left</i>	<i>Right</i>
<i>Up</i>	3 0	3 3
<i>Down</i>	0 5	5 0

... and if Row plays *Down* then Column's best response is *Right*;
so Column has a weakly dominant strategy of *Right*.

To make Row indifferent, Column plays *Right* w.p. $p = 5/8$, as in previous part.

NE is $(Up, Right \text{ w.p. } \geq 5/8)$.

3. Janet is an expected utility maximiser whose utility of wealth is $u(w) = \ln w$. Her initial wealth is 64, and she can invest in a risky project with a net return of +26 or -24, each equally likely. (So, if it succeeds, her final wealth will be 90, but if it fails, her final wealth will be 40.)

- (a) What is the expected net return of the project?

What is Janet's risk premium for the project?

Show that she won't undertake the project by herself.

She asks Sam (who has the same preferences and initial wealth as Janet) to share the risky project with her – they would each get a net return of either +13 or -12.

- (b) What is their individual expected net return and their risk premium for their share of the project?

Will the two of them accept the opportunity to share the project?

If not, how many similar people do they need?

Explain why sharing with more people makes the project more acceptable.

- (c) What size of group would maximise the expected utility of each member?

Why does sharing become less beneficial when the group gets bigger than this?

[25%]

- (a) expected net return $= \frac{1}{2} * (+26) + \frac{1}{2} * (-24) = 1$.

$$EU = \frac{1}{2} \ln(64 + 26) + \frac{1}{2} \ln(64 - 24) = \frac{1}{2} \ln(90) + \frac{1}{2} \ln(40) = \frac{1}{2} \ln(3600) = \ln(60).$$

certainty equivalent $= 60 < 64$ so she won't take on the project.

$$\text{risk premium } 64 + 1 - 60 = 5.$$

- (b) individual expected net return $= \frac{1}{2} * (+13) + \frac{1}{2} * (-12) = \frac{1}{2}$.

$$EU = \frac{1}{2} \ln(64 + 13) + \frac{1}{2} \ln(64 - 12) = \frac{1}{2} \ln(77) + \frac{1}{2} \ln(52) = \frac{1}{2} \ln(4004) \approx \ln(63.28).$$

certainty equivalent $\approx 63.28 < 64$ so they won't take on the project.

$$\text{risk premium} \approx 64 + \frac{1}{2} - 63.28 = 1.22.$$

Need n , s.t. $\ln(64) \leq \frac{1}{2} \ln(64 + 26/n) + \frac{1}{2} \ln(64 - 24/n)$,

$$64^2 \leq (64 + 26/n) * (64 - 24/n), \quad 0 \leq 64 * 2/n - 26 * 24/n^2, \quad n \geq 39/8, \quad \text{so } \boxed{n = 5}.$$

When the risk is shared with more people,

the risk premium initially falls faster than the expected net return.

Here, the individual expected net return $= 1, \frac{1}{2}, \frac{1}{3}, \dots, 1/n, \dots$,

and the risk premium $\approx 5, 5 * (\frac{1}{2})^2, 5 * (\frac{1}{3})^2, \dots, 5 * (1/n)^2, \dots$

- (c) $\max_n (64 * 2/n - 26 * 24/n^2), \quad n = 2 * 39/8, \quad \text{so } \boxed{n^* = 10}$. (SOC OK for max.)

For $n \geq 10$ the fall in the individual expected net return dominates the decrease in the already very small risk premium.

4. Three qualities of second-hand bicycles are available in equal numbers. There are many buyers and sellers; they have different values for each quality of bicycle, and everyone knows the value that each type of consumer attaches to each quality of bicycle.

Quality	Buyer's value	Seller's value
<i>High</i>	100	75
<i>Medium</i>	65	60
<i>Low</i>	30	45

- (a) What is the efficient outcome?

Buyers do not know the quality of any particular bicycle for sale, but sellers do know the quality of the one that they own. The price at which any bicycle is traded will be determined by supply and demand. No-one wants more than one bicycle.

- (b) What is the equilibrium outcome?

Now assume that the sellers can offer a reimbursement of 50 to the buyer, payable if the bicycle breaks down; those of high quality never fail, those of low quality always fail, and those of medium quality fail with probability $\frac{1}{2}$.

- (c) Show that there is an equilibrium with $85 \leq p < 95$ that is efficient.

- (d) Are there any other (inefficient) equilibria in which trade occurs?

[25%]

- (a) Efficient outcome: those who value the bike more own it.

High and *Medium* quality should be owned by the Buyers,
Low quality by the Sellers.

There should be no mention of prices, or of buying & selling.

- (b) If $p \geq 75$, all Sellers would sell, ave. value to Buyers = $65 < p$.

If $75 > p \geq 60$, S's would sell *Med.* & *Low* quality, ave. value to B's = $47\frac{1}{2} < p$.

If $60 > p \geq 45$, S's would sell *Low* quality, value to B's = $30 < p$.

No trade can be sustained at any price, which is the equilibrium outcome.

- (c) Introduction of the (expected) reimbursement increases the value of *Low* quality bikes by 50, and of *Med.* quality bikes by $\frac{1}{2} \cdot 50 = 25$.

Quality	Buyer's value	Seller's value
<i>High</i>	100	75
<i>Medium</i>	90	85
<i>Low</i>	80	95

If $p \geq 95$, all Sellers would sell, ave. value to Buyers = $90 < p$.

If $95 > p \geq 85$, S's would sell *High* & *Med.* quality, ave. value to B's = $95 > p$.

If $85 > p \geq 75$, S's would sell *High* quality, value to B's = $100 > p$.

So a price p with $95 > p \geq 85$ results in the efficient outcome from part (a).

- (d) So a price p with $85 > p \geq 75$ results in an equm. in which only *High* quality bikes are traded, which is inefficient.

Part B

10. Alex's has a house worth h and some financial assets worth f . Alex's house is in an area that is vulnerable to hurricanes; if there is a violent storm then the house will flood and be worth nothing, but the financial assets will be unaffected; if there is no flood the value of all her wealth will be unchanged. The probability of flooding is π .

Insurance against flood damage is available at a premium of p per unit of cover, so an amount of cover q costs pq .

However, $p > \pi$.

- (a) In a diagram illustrating wealth in the two states 'no flood', 'flood', mark the points indicating her final wealth if she buys no insurance, and if she buys full insurance. If Alex buys an amount of cover q (with $0 \leq q \leq h$), what is her final wealth in the two states 'no flood', 'flood'? Mark such a point on your diagram. [15%]

no insurance: $(f + h, f)$

full insurance: $(f + (1 - p)h, f + (1 - p)h)$

-- this is on the 45° line (the certainty line)

cover q : $(f + h - pq, f + (1 - p)q)$

-- this is on the straight line joining 'no insurance' & 'full insurance'

- (b) Alex is an expected utility maximiser, and has a utility function $u(\cdot)$ that is increasing and concave.

Does this mean that she is risk averse?

Write down her expected utility $E[u]$ if she buys an amount of cover q .

By examining the value of $\partial E[u]/\partial q$ when $q = h$, or otherwise, explain why Alex will choose to underinsure the house. [30%]

Yes: increasing and concave utility function implies Alex is risk averse.

$$EU[q] = (1 - \pi)u(f + h - pq) + \pi u(f + (1 - p)q)$$

$$\frac{\partial EU[q]}{\partial q} = -p(1 - \pi)u'(f + h - pq) + (1 - p)\pi u'(f + (1 - p)q)$$

$$\begin{aligned} \left. \frac{\partial EU[q]}{\partial q} \right|_{q=h} &= -p(1 - \pi)u'(f + h - ph) + (1 - p)\pi u'(f + (1 - p)h) \\ &= (\pi - p)u'(f + (1 - p)h) \\ &< 0, \text{ since } \pi < p \text{ and } u' > 0 \end{aligned}$$

EU is decreasing as $q \uparrow h$, therefore Alex would do better by stopping short of full insurance.

(c) Let $u(y) = \ln(y)$.

Assuming that Alex wants to buy *some* insurance, find q^* , the optimal amount of cover.

What condition, in terms of h , f , π and p , would imply that $q^* = 0$?

How does that relate to the slope of Alex's indifference curve at the endowment?

[30%]

$$\text{FOC: } -p(1-\pi)u'(f+h-ph) + (1-p)\pi u'(f+(1-p)h) = 0$$

When $u(y) = \ln(y)$:

$$\frac{-p(1-\pi)}{f+h-pq} + \frac{(1-p)\pi}{f+(1-p)q} = 0$$

$$p(1-\pi)(f+(1-p)q) = (1-p)\pi(f+h-pq)$$

...

$$q^* = \boxed{\frac{(\pi-p)f + (1-p)\pi h}{p(1-p)}}$$

$$q^* = 0 \Leftrightarrow (\pi-p)f + (1-p)\pi h = 0$$

When this condition holds, the slope of the indifference curve at the endowment is the same as the slope of the line between no insurance & full insurance, i.e. the indifference curve is tangent to the (unfair) odds line at the endowment.

(If 'negative insurance' is not allowed, then $q^* = 0 \Leftrightarrow (\pi-p)f + (1-p)\pi h \leq 0$; the condition on the slope becomes 'steeper than' instead of 'the same as'.)

(d) When $q^* > 0$, how would it change if

(i) the value of Alex's house increased,

(ii) the value of Alex's financial assets increased?

Comment on each of these changes.

[25%]

$$\frac{\partial q^*}{\partial h} = \frac{\pi}{p} : \text{ this is positive but } < 1;$$

when Alex's house become more valuable, she takes out more insurance -- but not proportionately so, because the rate of cover is unfavourable.

$$\frac{\partial q^*}{\partial f} = \frac{\pi-p}{p(1-p)} : \text{ this is negative;}$$

someone with $\ln$ utility has Constant Relative Risk Aversion,

and therefore Decreasing Absolute Risk Aversion --

when Alex is wealthier she takes more risks and so buys less insurance.